Edge Feature Enhancement for Fine-Grained Segmentation of Remote Sensing Images

Zhenxiang Chen, Tingfa Xu, Yongzhuo Pan, Ning Shen, Huan Chen, and Jianan Li

***Abstract*— Fine-grained segmentation of remote sensing mineral images plays a crucial role in the investigation and monitoring of mineral resource. In deep-learning methods, fine-grained segmentation and edge detection are closely related in both data construction and feature extraction. However, the open-pit mineral areas in remote sensing images are heavily affected by complex natural environmental interference, posing challenges for precise data annotation and dataset construction. In view of this, we introduce the fine-annotated remote sensing mineral image (Fine-RSMI) dataset, which includes a total of 10 225 images with finely annotated edges, while also introducing challenges such as multiscale and edge irregularities. To tackle the challenge of fine-grained segmentation in irregular edges, we propose a hierarchical fusion edge feature enhancement framework. Our framework consists of an edge detail feature enhancement module (EDFEM) and an edge supervision module (ESM). EDFEM vertically cascades multiple feature fusion units to obtain high-order complementary information for refining edge features. ESM further supervises network reinforcement learning of mineral area edges using ground truth edge maps to improve edge segmentation performance. Both modules work in a plug-and-play manner, enabling effortless integration into existing segmentation networks. Our method achieves further performance improvement in many general remote sensing segmentation frameworks, reaching the best results of 74.12% mean intersection-over-union (mIoU) on Fine-RSMI dataset and 78.64% mean accuracy (mAcc) on WHDLD dataset. Fine-RSMI dataset and code will be available at https://github.com/chenmu1204/czx.**

***Index Terms*— Edge feature enhancement, fine-grained segmentation, open-pit mineral area, remote sensor.**

I. Introduction

THE accurate segmentation and assessment of remote sensing imagery of open-pit mineral resources are crucial for resource management [1], [2] and environmental monitoring [3]. To achieve high performance in the aforementioned applications, a robust, high-precision, and fast automatic classification method is required. Deep learning-based [4], [5] remote sensing image segmentation is a cutting-edge approach, and the fine-grained segmentation of mineral area can provide more detailed information for decision-making and management in this field. This not only supports the efficient and sustainable exploitation of mineral resources but also promotes broader environmental protection and the rational use of mineral resources. However, the complex terrain, dynamic changes, and noise in mineral remote sensing imagery present significant challenges to data processing, data construction, and fine-grained segmentation.

Deep-learning methods for segmenting mineral areas require a substantial amount of high-quality data-driven inputs. Nonetheless, this domain faces significant challenges due to the sparse availability of data and the generally coarse annotations within existing datasets. As illustrated in Fig. 1(a) and (c), mineral areas display complex local features with gradual transitions between the mineral areas and their surrounding environments. Existing dataset [6] available for mineral areas is typically annotated with relatively coarse granularity at their edges, which complicates the task of achieving accurate segmentation. Furthermore, open-pit mineral areas are often situated in mountainous regions or plains and can easily be confused with nearby vegetation, roads, buildings, waste heaps, and land cultivation. The annotation of such datasets requires considerable expertise and domain-specific knowledge.

Current datasets do not suffice for detailed and finer pixel-level classification tasks. To support the advancement of data-driven fine-grained segmentation frameworks, we introduce the finely annotated dataset fine-annotated remote sensing mineral image (Fine-RSMI). This dataset aims to establish a benchmark for the fine-grained segmentation of remote sensing mineral images. It comprises meticulously annotated 10 225 images from satellite captures featuring diverse open-pit mineral area scales and notably irregular edges, as depicted in Fig. 1(d). Additionally, with the inclusion of 22 sub-classes of mineral areas, it presents significant challenges due to substantial intraclass variations and pronounced interclass similarities.

The presence of multiscale and irregular edges significantly complicates model's ability to discern target boundaries. And the prevalent intraclass inconsistency and interclass ambiguity contribute to a higher likelihood of misclassification, particularly for pixels near the edges of mineral areas. To address

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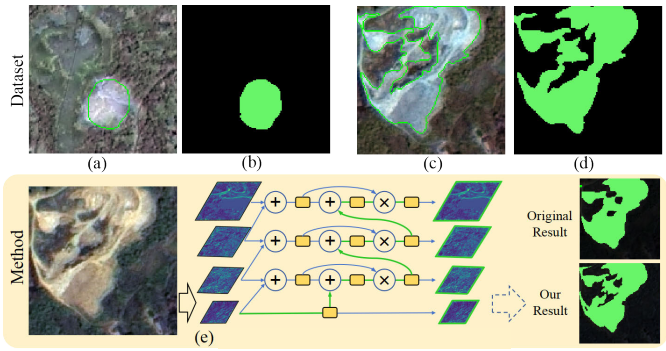


Fig. 1. Motivation of our method. (a) Existing dataset RSMI [6] examples of mineral areas, green color represents the annotated edges of mineral areas. (b) Ground truth corresponding to (a). (c) Examples of Fine-RSMI dataset exhibits irregular mineral area edges, and existing gradual transition between mineral area edges and background. (d) Ground truth corresponding to (c). (e) EDFEM adopts vertical cascading and hierarchical fusion of adjacent level feature maps to obtain edge detail features with prominent edge information. Under such a design, our result performs better at complex edges.

this challenge, we propose a novel framework: a vertical cascading and hierarchical fusion framework for edge feature enhancement. This framework is designed to improve the precision of edge pixel classification in mineral areas by systematically enhancing edge-delineation capabilities.

Our framework is specifically engineered to augment feature representation at the edges of mineral areas. It integrates the transformer's capacity to model long-range dependencies [7] with the convolutional neural network's (CNNs) ability to fuse multilevel features, thereby creating an innovative method for feature enhancement. Open-pit mineral areas in remote sensing images typically include large, connected regions with a wide range of scales and irregular edges. Therefore, improving pixel segmentation accuracy within local edge areas is essential for overall performance. Our framework includes two primary components: edge detail feature enhancement module (EDFEM), which improves edge feature extraction, and edge supervision module (ESM), which bolsters edge learning through the application of an edge supervision loss.

As illustrated in Fig. 1(e), EDFEM employs a vertical cascading approach to integrate adjacent hierarchical feature maps, thereby extracting detailed features enriched with prominent edge information. The primary data flow, represented by green arrows in the figure, integrates high-order complementary information from multiple feature hierarchies. This integration serves as an auxiliary mechanism, ensuring that the fused features embody rich multilevel detail and semantic content. For a more intuitive understanding, we conceptualize the fused feature maps as amplifiers, each assigned different weights. These amplifiers enhance the contrast between edge and nonedge regions during the network's layer-wise upsampling process, thus accentuating the edges of targets.

Additionally, the ESM guides the EDFEM in calculating the edge loss between the ground truth edges and the edge feature maps. This guidance employs principles of reinforcement learning to strengthen pixel features within the local edge regions. Consequently, the synergistic operation of EDFEM and ESM significantly enhances the network's edge recognition capabilities, leading to improved accuracy in pixel classification at these critical junctures. This method improves performance without increasing the cost of image inference. The effectiveness of our method over traditional methods is demonstrated on the right side of Fig. 1(e), where a comparison of segmentation results is presented.

The proposed modules can be seamlessly integrated into existing segmentation frameworks to enhance their predictive performance. Our method consistently improves performance, not only on our Fine-RSMI dataset but also on the publicly available WHDLD [8] and ISPRS [9] datasets, thereby underscoring its flexibility and versatility.

The main contributions of this work are as follows.

1) We establish Fine-RSMI Dataset, a crucial resource for benchmarking and advancing research in remote sensing images fine-grained segmentation.
2) We propose a hierarchical fusion edge feature enhancement framework, comprising EDFEM and ESM. EDFEM enhances the network's edge feature capture capability, while ESM collaborates with EDFEM to collectively instruct the network in discerning edge details.
3) EDFEM and ESM operate in a plug-and-play manner, seamlessly integrating into existing frameworks to enhance the segmentation performance of remote sensing images.

II. Related Work

A. Datasets for Remote Sensing Mineral Images

Existing remote sensing segmentation datasets are predominantly tailored for land use (WHDLD [8], GID [10], DeepGlobe 2018 [11]), and urban scenes (Massachusetts Building Dataset [12], ISPRS Vaihingen,and Potsdam Dataset [9]), characterized by a single data source and scale, with annotations generally being coarse. Remote sensing images inherently feature large imaging scales, extensive coverage, and complex, variable background information, which collectively pose significant challenges to fine annotation. Currently, the datasets suitable for fine-grained segmentation of open-pit mineral areas are scarce. The RSMI Dataset [6] is applied to open-pit mineral areas. It exemplifies these issues, as its edge annotations lack the precision required to fulfill the stringent data needs of fine-grained segmentation.

The uneven distribution of mineral area types, coarse labeling practices, and the insufficient diversity of depicted mineral areas are significant obstacles that limit advancements in this field. These factors complicate accurate modeling and understanding of mineral area characteristics. To address these challenges, we develop the Fine-RSMI dataset, which not only provides finer annotations at the edges but also introduces complexities such as multiscale features, internal voids, and edge irregularities, specifically tailored for the fine-grained segmentation of mineral areas.

B. Fine-Grained Segmentation of Remote Sensing Images

Fine-grained segmentation of remote sensing images addresses critical issues such as edge pixel-level accuracy, complex backgrounds, and multiscale variations. To improve

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pixel-level accuracy, integration of edge features with semantic segmentation is essential.

In CNN-based methods, techniques such as multiscale processing, dilated convolutions, transpose convolutions, skip connections, and attention mechanisms are usually employed to achieve superior performance. CEN [13] attempts to incorporate edge information into CNN models, enabling the model to better identify edge details. Transformer models [7], [14] are inherently suited for sequence modeling, and they treat image pixels as sequences, which enables more effective modeling of fine texture information, crucial for tasks involving fine-grained segmentation. There are some tricks that can further enhance fine-grained segmentation. One solution is to carry detailed information in a hierarchical encoding and decoding [15] or multi branch manner [16]. Such as DCFAM in DC-Swin [17] adopts densely connected feature aggregation method, but our continuous vertical cascading approach offers more effective feature extraction capabilities for the edges of irregular targets. Another method is to overlay detailed information during the feature extraction process [18]. Some auxiliary methods [19] such as edge guidance and loss functions constraint are also often applied.

The above methods primarily address pseudo-edges that manifest as gradient changes. Our method similarly concentrates on target edges but narrows its focus to pixels within the local vicinity of these edges. Moreover, during feature extraction, we guide the network to intensify the learning of edge pixels using reinforcement learning strategies. This approach encourages the model to establish clearer decision boundaries for classification, enhancing the accuracy and effectiveness.

C. Methods for Optimizing Edge Processing

In conventional semantic segmentation methods, the quality of segmentation at edges is a crucial performance metric. Conventional methods improve the segmentation accuracy of edge pixels by edge detection algorithms (e.g., Canny and Sobel) or specially designed networks. IBCO-Net [20] simultaneously considers integrity, edge, and corner information, proposing a three-stage network to integrating them. PEGNet [21] treats edges as symbols reflecting erroneous pixels and utilizes them to guide feature extraction for semantic segmentation. CEN [13] divides network into two branches, edge detection and semantic segmentation. They then merge the outputs to obtain features with abundant edge information. EdgeDG [22] also focus on edge feature extraction, but usually specialize in designing complex structures.

In general, the existing edge optimization techniques can be summarized into two aspects, one is to improve the segmentation accuracy of the edge pixels themselves, and the other is to enhance the correlation between the edge and the target. In this work, taking these two aspects into account, we design the feature enhancement module and the optimized loss function as plug-and-play modules, which show significant advantages in improving the segmentation performance of remote sensing images. These improvements not only enhance the segmentation accuracy and robustness, but also greatly simplify the application of the method, reflecting high practicality and popularization value.

TABLE I
INFORMATION ABOUT SATELLITES USED FOR COLLECTING DATA

Satellite	Resolution	Spectra	Orbit	Launch year
Gaofen 1	2 m	R, G, B	645 km	2013
Gaofen 2	0.8 m	R, G, B	631 km	2014
Gaofen 6	2 m	R, G, B	634 km	2018
Resource 3	2 m	R, G, B	505 km	2012

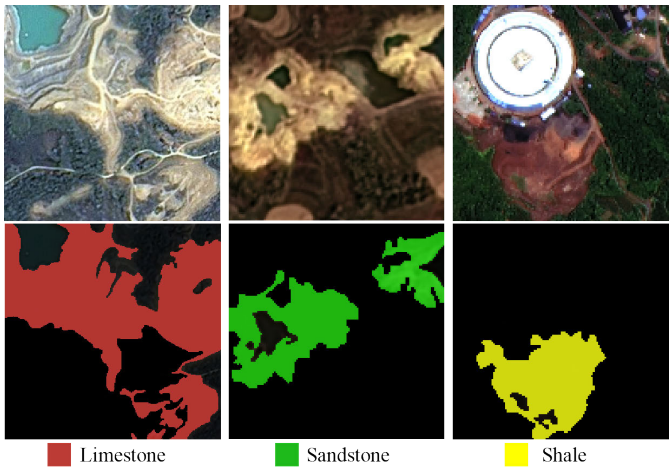


Fig. 2. Pixel level annotation labels for Fine-RSMI dataset, with gradient irregular edges and internal voids. The red, green, and yellow labels represent typical open-pit mineral areas located in limestone, sandstone, and shale.

III. FINE-RSMI DATASET BENCHMARK

Remote sensing mineral area datasets are relatively scarce compared to datasets in other domains, especially high-quality finely annotated data specific to certain types of minerals. To bridge this gap, we establish the Fine-RSMI.

A. Dataset Construction

1) Image Sources and Characteristics: Our data are sourced from China's Gaofen series of satellites: Gaofen 1, Gaofen 2, Gaofen 6, and Resource 3, which collect various open-pit mineral areas in the Chongqing mountain area. Basic parameters of the satellites used are shown in Table I. Data collection period is from March to December every year after 2018, and the weather conditions and solar altitude are randomly captured, increasing the richness of the data.

Mineral images are characterized by complex terrain and texture, high dynamic changes, and much noise and interference. Mineral resource boundaries are inherently elusive, often blurred by activities that involve Earth disturbance, complicating the delineation process. These characteristics increase the complexity of data processing. Therefore, extensive manpower is required for field validation to ensure the accuracy of annotated samples. As the data scale expands and the types of mineral types are further subdivided, we will no longer require exploration personnel for rough classification. However, professional experts will still be needed to screen and analyze the final results of the model segmentation.

2) Label Annotation: We meticulously annotate the edges of each mineral area at the pixel level. Following visual interpretation by remote sensing geological experts, we convert

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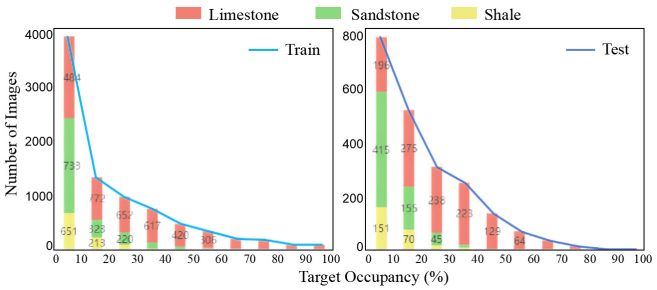


Fig. 3. Statistics of Fine-RSMI dataset. (Left) Distribution of images corresponding to the ratio of mineral area pixels to the total number of pixels in the training set. (Right) Distribution of images corresponding to the ratio of mineral area pixels to the total number of pixels in the testing set.

TABLE II
NUMBER OF IMAGES IN DIFFERENT CLASSES

Class	Limestone	Sandstone	Shale	Total
Train	4765	2443	956	8164
Test	1173	626	239	2038

mineral maps in remote sensing images into vector maps representing minerals. Subsequently, we utilize the ArcGIS vector-to-grid tool to generate annotated images. To ensure the precision of mineral spot areas classification, we initially engage remote sensing geological experts in annotation and differentiation tasks, followed by comprehensive field verification of all identified spots. After filtering out misclassified mineral spot areas, the remaining spot areas are confirmed to be accurately classified.

Fig. 2 displays visualizations of partial pixel-level labeled images. The outlined open-pit mineral areas exhibit extreme irregularity, presenting challenges for high-quality fine annotation. These images have more refined edge annotations than those shown in Fig. 1(a) of the mineral area dataset. Fine annotation addresses the issue of inaccurate classification caused by blurred semantic information at the edges of open-pit mineral areas. This facilitates more precise pixel-level segmentation and simultaneously simplifies both data processing and framework design.

B. Data Statistics

Fine-RSMI dataset contains 10 225 meticulously annotated remote sensing mineral images. Among these, there are 6417 images with a resolution of 128 × 128 and 3808 images with a resolution of 256 × 256. We partition the dataset into 8175 images for training and 2050 images for testing. Table II delineates the quantity information of mineral area images, wherein the actual number of target images is fewer than the total due to the presence of some negative samples and some images have multiple mineral areas.

Fine-RSMI dataset encompasses three major classes of minerals and 22 subclasses.

1) *Limestone Group:* Solvent-grade limestone, metallurgical-grade dolomite, limestone, cement-grade limestone, building stone-grade limestone, decorative stone-grade limestone, and quicklime-grade limestone.

TABLE III
ATTRIBUTE ANALYSIS OF REMOTE SENSING SEGMENTATION DATASETS

Dataset	Scene	Scale Variation	Complex Edge	Internal Voids
WHDLD [8]	Land use	✓	✓	✓
GID [10]	Fine land cover	✗	✓	✓
DeepGlobe [11]	Rural area	✗	✓	✓
Massachusetts [12]	Urban area	✗	✗	✓
ISPRS [9]	Urban area	✗	✗	✓
RSMI [6]	Mineral resource	✗	✗	✗
Fine-RSMI	Mineral resource	✓	✓	✓

2) *Sandstone Group:* Quartzite, metallurgical-grade sandstone, metallurgical-grade quartzite, glass-grade quartzite, Sandstone, glass-grade sandstone, aggregate-grade sandstone, ceramic-grade sandstone, construction-grade sandstone, natural quartz sand, construction-grade sand, and aggregate-grade sand.
3) *Shale Group:* Shale, brick, and tile-grade shale, and aggregate-grade shale.

Fig. 3 analyzes the distribution of target area pixel proportions in the Fine-RSMI training and testing sets. From the statistical graphs, it can be observed that the distribution of mineral area pixel proportions ranges from 2% to 100%, reflecting a rich range of scale variation. Notably, the ratio distributions of the training and testing sets are consistent, and the quantities of each class are comparable.

C. Comparison With Remote Sensing Segmentation Datasets

Remote sensing segmentation datasets are commonly applied to urban scenes, rural scenes, and land use. Table III compares the attributes of several popular remote sensing segmentation datasets. Most datasets do not exhibit the large-scale variation of 2%–100% like Fine-RSMI and WHDLD. The complex terrain and dynamic changes in open-pit mineral areas require segmentation methods that can handle high noise, complex backgrounds, and complex morphology. Urban areas, on the other hand, rely more on rich data and mature models, with clearly distinguishable targets and backgrounds and relatively standardized processes. These differences highlight the importance of customizing segmentation methods to specific application scenarios.

In conclusion, Fine-RSMI dataset establishes a benchmark for mineral resource scenes, effectively filling a notable gap in this field. It introduces unique features related to multiscale variations, edge irregularity, and the presence of internal voids, thereby setting a new standard for fine-grained segmentation in open-pit mineral areas.

IV. METHOD

A. Overview

Our framework consists of three main parts: transformer encoder–decoder, edge detail feature enhanced module (EDFEM) and ESM, as shown in Fig. 4. Transformer-based encoder extracts global semantic information from remote sensing images through a sequence of consecutive transformer blocks. EDFEM enhances the detailed texture information

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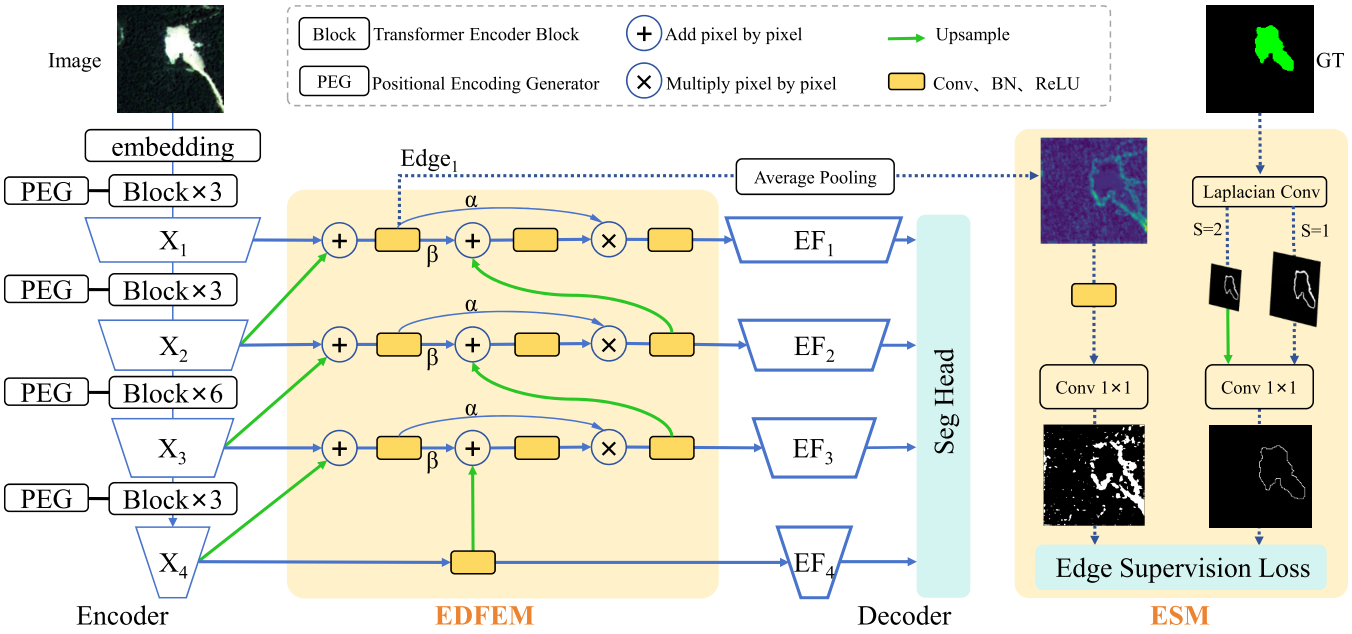


Fig. 4. Overall framework of our fine-grained segmentation network, mainly consisting of transformer encoder–decoder, EDFEM, and ESM. X_1, X_2, X_3, and X_4 represent the original hierarchical features, EF_1, EF_2, EF_3, and EF_4 represent the corresponding enhanced features, α is edge weight coefficient, and β is feature weight coefficient.

at the edges of images to obtain finer pixel-level feature representations. ESM conducts supervised training on intricate target edges to further enhance segmentation accuracy at the target boundaries.

B. Preliminaries

We chose Twins-PCPVT [7] as the backbone of our framework, which inherits the advantages of PVT [23] that outputs feature pyramids, providing a foundation for us to propose layer-by-layer feature enhancement methods.

1) Twins-PCPVT Backbone: As shown in Fig. 4, Twins-PCPVT [7] backbone consists of patch embedding and four consecutive downsampling stages. Each stage contains a different number of transformer blocks and a positional encoding generator (PEG) [24]. In our framework, patch size is 4×4, so the feature dimension of each patch is $4 \times 4 \times 3 = 48$. The original features are directly inputted into the encoding block, with the conditional position coding generator placed after the first encoding block of each stage to generate conditional position coding. The output feature map is obtained by adding pixel-level details to the PEG processed features through a straightforward skip connection.

2) Positional Encoding Generator: Detailed PEG is shown in PVT [24]. The simplest form of using PEG is to dynamically generate position encoding based on the local neighborhood of the input token without the need for batch normalization of 2-D deep convolution. First, we reshape the flattened sequence $\boldsymbol{X} \in \mathbb{R}^{N\times C}$ to $\boldsymbol{X}' \in \mathbb{R}^{H\times W\times C}$, and repeatedly apply a function (various types of convolutions or other forms of functions) on the local patch of $\boldsymbol{X}'$ to generate conditional positional encoding $\boldsymbol{E} \in \mathbb{R}^{H\times W\times C}$. Finally, we reshape the features back to the size $\boldsymbol{X}'' \in \mathbb{R}^{N\times C}$. PEG can be effectively achieved through a zero filled 2-D convolution.

3) Transformer Block: Two consecutive Twins transformer blocks [7] with locally grouped self-attention (**LSA**) and global subsampled attention (**GSA**) can be represented as

$$\hat{z}^l = \mathbf{LSA}(\mathbf{LN}(z^{l-1})) + z^{l-1} \tag{1}$$

$$z^l = \mathbf{FFN}(\mathbf{LN}(\hat{z}^l)) + \hat{z}^l \tag{2}$$

$$\hat{z}^{l+1} = \mathbf{GSA}(\mathbf{LN}(z^l)) + z^l \tag{3}$$

$$z^{l+1} = \mathbf{FFN}(\mathbf{LN}(\hat{z}^{l+1})) + \hat{z}^{l+1} \tag{4}$$

where l is layer index, z^{l-1} represents initial input feature of transformer block, $\hat{z}^l$ represents fused feature after **LSA** block, z^l represents fused feature after the first **FFN** layer, $\hat{z}^{l+1}$ represents fused feature after **GSA** block, and z^{l+1} is the final output feature of transformer block. **LN** represents layer normalization, and **FFN** represents feed forward layer.

Transformer blocks for the four stages in the entire backbone network are (3, 3, 6, 3). The image with the original resolution of $H \times W$ becomes $(H/4) \times (W/4)$ after stage 1, subsequent layer-by-layer downsampling to $(H/32) \times (W/32)$. The output feature dimensions are (64, 128, 320, 512). The generated hierarchical feature pyramid has different levels of delicacy information at different resolutions, and the shallowest features have the richest detailed texture information, which plays a crucial role in fine-grained segmentation of remote sensing images.

4) Upperhead Design: For the fine-grained segmentation of remote sensing images, the texture information and semantic information of the target are equally important. Therefore, it is necessary to fully utilize features of different resolutions to achieve better segmentation performance.

Upperhead mainly consists of pyramid pooling module (PPM), cascading addition module, and fusion module. PPM uses global pooling with different pooling scales to obtain information on different receptive fields, enabling the model to

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establish local and global connections, thereby enhancing the model's discriminative ability for different classes. To further enhance the expression ability of detailed information, we first unify the number of channels in the hierarchical feature map to C before feature fusion. Then, we extract edge features to enhance the feature expression of edge information, and assist the feature enhancement module in achieving finer pixel-level segmentation at the edges.

C. Edge Detail Feature Enhanced Module

EDFEM specifically enhances edge detail information and can flexibly integrate into other segmentation frameworks with feature pyramid structure, offering a plug-and-play functionality. As mentioned earlier, userhead directly integrates features from different levels in the decoder through element-wise addition and cascading. This results in a large amount of redundant information, weakening the complementarity between features at different levels. Consequently, it leads to inaccurate positioning and misclassification of local pixels at the edges. We take the high-level semantic information at the bottom as the main focus, and use feature maps that integrate different levels of detail information as auxiliary. This strategy gradually strengthens the edge detail information of the bottom features, promoting the network's learning of target edges.

1) Edge Information Extraction: We use $\boldsymbol{X}_i$ and $\boldsymbol{X}_{i+1}$ to represent adjacent-level feature maps, unifying the number of channels to C, which can reduce the number of parameters for subsequent operations, and using ReLU for activation. We define **Edge** to guide network training

$$\mathbf{Edge}_i = \boldsymbol{f}_1(\boldsymbol{X}_{i+1} + \boldsymbol{X}_i) \tag{5}$$

where $\boldsymbol{f}_i$ represents performing 1×1 Conv, batch normalization, and ReLu operations sequentially of ith layer. $\mathbf{Edge}_i$ represents the edge detail feature of the ith layer. After upsampling, $\boldsymbol{X}_{i+1}$ is added pixel by pixel with $\boldsymbol{X}_i$ to achieve simple multiscale edge detail feature fusion.

2) Enhanced Feature: We define $\boldsymbol{EF}_i$ as the enhanced-feature of the ith layer

$$\boldsymbol{EF}_i = \boldsymbol{f}_3(\alpha\mathbf{Edge}_i \cdot \boldsymbol{f}_2(\boldsymbol{EF}_{i+1} + \beta\mathbf{Edge}_i)) \tag{6}$$

where the number of channels in the input and output layers of $\boldsymbol{f}_1$ and $\boldsymbol{f}_2$ is both C. In the network structure of Twins-PCPVT [7], the number of channels for f_3 is from C to $\{\mathrm{C_i} \mid \mathrm{C}_1, \mathrm{C}_2, \mathrm{C}_3 = 64, 128, 320\}$, α, and β is corresponding weight coefficient.

As shown in Fig. 4, in order to obtain high-order complementary information across multiple feature levels, we concatenate multiple units vertically and calculate the detailed features of different layers and receptive fields. Specifically, edge maps characterizing detailed texture information exhibit more prominent features at the target edge regions. After weakening by the coefficients α and β, they still carry significant weight at the edges of the mineral areas. Initially, we perform pixel-wise addition with features from the previous layer, followed by scaling with a factor of $\alpha\mathbf{Edge}_i$, resulting in a differential feature representation between the smooth region within the mineral area and its edges. Simultaneously, the fused features possess multilevel detail and semantic information. We design α and β as adjustable parameters, limited to a smaller range of values less than 1 to avoid excessive masking of advanced semantic information.

3) Plug-and-Play Working Manner: EDFEM and ESM are plug-and-play modules that can be seamlessly integrated into backbones with multilevel features. It is expressed by the following equation:

$$\boldsymbol{X}_i' = \mathbf{EM}(\boldsymbol{X}_i) \tag{7}$$

where **EM** denotes the function encompassing operations of both the EDFEM and ESM modules, $\boldsymbol{X}_i \in \mathbb{R}^{B\times \mathrm{C}\times \mathrm{H}\times \mathrm{W}}$ represents the original feature maps extracted by any backbone, and $\boldsymbol{X}_i' \in \mathbb{R}^{B\times \mathrm{C}\times \mathrm{H}\times \mathrm{W}}$ represents the enhanced feature maps after passing through our plug-and-play modules.

D. Edge Supervision Module

We use the ground truth edges as the guidance of detail feature map to guide the low-level layers to learn the feature of edge details. This branch is discarded in the inference phase. Therefore, this side-information can easily boost the accuracy of segmentation task without any cost in inference.

1) Detail Feature Map and GT Edge Map: As illustrated in Fig. 4, the edge spatial detail feature map $\mathbf{Edge}_1$, serves as one input for ESM after passing through average pooling layer. In another branch, we generate binary edge maps from ground truth labels as another input for loss computation. This operation consists of Laplacian convolutions with different strides (1, 2) and a trainable 1×1 convolution. By setting an appropriate binary classification threshold, accurate ground truth edge maps can be generated. Subsequently, we employ ground truth guidance to facilitate the learning process of edge features, explicitly enhancing the network's performance at the edges of mineral areas.

2) Edge Supervision Loss Function: In edge loss computation, the number of edge pixels is significantly smaller than nonedge regions, resulting in a significant class imbalance for binary classification problems where edge values are 1 and nonedge values are 0. To address this issue, PEGNet [21] employs region-sensitive loss functions to compute confidence for edge and nonedge pixels separately. Here, considering the superior performance of dice loss, we jointly optimize detail learning using binary cross-entropy and dice loss. Dice loss is insensitive to the quantity of foreground pixels and only computes overlap, which helps alleviate the class imbalance problem. The total loss function is represented as

$$\mathbf{Loss} = \mathbf{L}_{\mathrm{seg}} + (\lambda_1\mathbf{L}_{\mathrm{bce}} + \lambda_2\mathbf{L}_{\mathrm{dice}}) \tag{8}$$

where $\mathbf{L}_{\mathrm{seg}}$ represents segmentation loss, and the remaining part is edge supervision loss. $\mathbf{L}_{\mathrm{bce}}$ represents the binary cross-entropy loss while $\mathbf{L}_{\mathrm{dice}}$ represents the dice loss, given as follows:

$$\mathbf{L}_{\mathrm{bce}}(p, g) = -\textstyle\sum_i^{H\times W}(p_i \times \log(g_i) + (1 - p_i) \times \log(1 - g_i)) \tag{9}$$

$$\mathbf{L}_{\mathrm{dice}}(p, g) = 1 - \frac{2\sum_i^{H\times W} p_i g_i + \epsilon}{\sum_i^{H\times W} p_i^2 + \sum_i^{H\times W} g_i^2 + \epsilon} \tag{10}$$

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where $H \times W$ is the size of the feature map, p_i denotes the predicted value of the ith pixel, and g_i denotes the ground truth value of the ith pixel. ϵ is a Laplace smoothing item to avoid zero division. In this article, we set $\lambda_1 = \lambda_2 = 0.5$.

V. Experiments

A. Implementation Details

Extensive experiments are conducted to verify the effectiveness of the proposed method on Fine-RSMI, WHDLD [8], ISPRS Vaihingen, and Potsdam [9] datasets. Fine-RSMI dataset is our customized dataset, as described in Section III. WHDLD [8] is a dense remote sensing dataset used for multilabel semantic segmentation tasks in geology. ISPRS Vaihingen and Potsdam [9] dataset division methodology follows the UNetFormer [25].

Our network is implemented using the Twins-PCPVT [7] framework, optimizing with AdamW. The learning rate is initialized as 6×10^{-5}. Images are resized to a spatial resolution of 512 × 512 during training and testing. All experiments are implemented on a single NVIDIA GeForce RTX 2080 Ti 12-GB GPU. The batch size is set to 4. We employ the mean intersection-over-union (mIoU) and mean accuracy (mAcc) across each class, and overall accuracy (OA) across each pixel as the evaluation metrics.

B. Feasibility Validation of Fine-RSMI Dataset

Our Fine-RSMI dataset is evaluated across a wide range of frameworks. As shown in Table IV, the upper parts are transformer-based frameworks, while the lower parts are CNN-based frameworks. Fine-RSMI dataset achieves an average performance of 94.46% OA, 65.09% mIoU, and 73.91% mAcc across different baseline frameworks.

Overall, transformer-based frameworks generally outperformed CNN-based methods. In particular, Twins [7], Swin [26], and DC-Swin [17] reach over 70% mIoU and 78% mAcc. This can be attributed to the architectural design of transformers, which utilize self-attention mechanisms to better capture long-range dependencies and global semantic information in remote sensing images, effectively addressing the specific tasks in our dataset. Specifically, Twins [7] achieves the best performance. We ultimately select Twins [7] as our baseline model.

TABLE IV
Performances of Transformers and CNNs on Fine-RSMI

Model	OA	mIoU	mAcc	Time/ms	Params/M
Twins [7]	**95.92**	**73.19**	**80.45**	178.7	53.23
Swin [26]	95.57	71.74	79.61	128.3	58.94
DC-Swin [17]	94.82	70.97	78.83	190.6	66.9
Segmenter [27]	95.36	69.54	77.10	87.0	40.72
PoolFormer [28]	95.44	69.39	77.55	122.2	15.65
SegFormer [29]	95.23	68.51	76.44	82.1	3.72
UNetFormer [25]	93.06	64.36	75.22	131.9	11.7
DPT [30]	93.84	62.77	74.90	225.1	110
BEiT [31]	94.02	62.48	72.66	273.8	161
ViT [14]	93.82	57.50	65.39	271.4	142
K-Net [32]	95.45	69.40	77.32	46.5	60.34
DeepLabV3 [33]	95.02	69.22	77.15	66.9	65.74
PointRend [34]	94.44	65.06	73.36	92.6	44.81
DeepLabV3+ [35]	94.83	63.86	71.96	106.4	41.21
SegNext [36]	93.94	62.96	73.01	21.4	4.23
BiSeNetV1 [37]	93.82	59.36	69.23	33.5	13.27
STDC [38]	93.60	57.68	65.75	**10.7**	8.27
BiSeNetV2 [39]	92.07	53.66	64.43	10.8	**3.36**

C. Results on Fine-RSMI Dataset

1) Main Results: We integrate our plug-and-play modules (EDFEM and ESM) into some frameworks illustrated in Table IV and present the corresponding experimental results in Table V. The symbol "+" denotes the framework after inserting both EDFEM and ESM into the original baselines. For instance, Twins+ indicates the framework using Twins [7] and incorporating our EDFEM and ESM. The incremental values in parentheses of Table V represent the quantitative improvement relative to the performance indicated in Table IV.

The results in Table V reveal that, across most methods, segmentation accuracy of the "Sandstone" class is generally lower, while the segmentation accuracy of other classes is relatively comparable and consistent. This observation can be attributed to two main factors: first, the imbalance in the number of samples among different classes in the dataset, and second, the large intraclass differences in "Sandstone" class mineral areas. As described in Section III, the "Sandstone" group comprises 11 subclasses of mineral areas, exhibiting significant differences in illumination, color, and shape.

Across all frameworks, the performance difference before and after integrating our modules is mainly attributed to the accuracy of the "Sandstone" class. The advantage of transformer architecture lies in modeling long-range dependencies. And our design focuses on enhancing the details information of the target. This indicates that the model's ability to model long-range dependencies is strengthened to some extent after incorporating our modules, alleviating the problem of large intraclass differences.

Additionally, Twins+ and Swin+ achieve outstanding performance with mIoU of 74.12% and 73.33%. Swin+ has the same backbone as DC-Swin [17] in Table IV, compared to which our Swin+ has a 2.36% mIoU improvement. Our approach achieves more effective performance improvement.

2) Efficiency Analysis: Among the seven frameworks presented in the experiments, the insertion of our modules results in significant improvements. Overall, the mIoU of the seven methods increased by an average of 1.19%, and the mAcc increased by an average of 1.11%. A series of experimental results demonstrate that our designed modules not only achieve the best performance on Fine-RSMI dataset but also lead to further performance enhancements across different frameworks. Table V compares the computational cost and runtime before and after incorporating our modules. We find that the additional module's parameters account for only about 3% of the original parameters. This indicates that our method maintains a good balance between computational cost and performance improvement.

3) Qualitative Results: Fig. 5 provides qualitative visual results. The results shown in the first row almost ignore the internal voids of the mineral areas, which are common in

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TABLE V
RESULTS OF SOTAS WITH OUR MODULES ON FINE-RSMI DATASET

Model	Time/ms	Params/M	OA	mIoU	mAcc	Background		Limestone		Sandstone		Shale	
						IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Twins+ [7]	201.9(+23.2)	54.81(+1.58)	**96.02**(↑ 0.10)	**74.12**(↑ 0.93)	**81.57**(↑ 1.12)	**96.16**	98.15	**79.27**	**89.12**	**56.05**	**66.99**	**65.01**	72.03
Swin+ [26]	135.8(+7.5)	60.52(+1.58)	95.83(↑ 0.26)	73.33(↑ 1.59)	81.56(↑ 1.95)	95.95	98.06	78.52	88.39	54.43	65.77	64.41	**74.00**
Segmenter+ [27]	93.7(+6.7)	41.61(+0.89)	95.54(↑ 0.18)	72.07(↑ 2.53)	78.62(↑ 1.52)	95.54	**98.41**	77.19	85.35	50.92	60.54	64.63	70.16
PoolFormer+ [28]	128.5(+6.3)	16.05(+1.60)	95.46(↑ 0.02)	70.47(↑ 1.08)	78.45(↑ 0.90)	95.64	98.10	76.80	86.94	45.58	55.19	63.86	73.58
K-Net+ [32]	69.6(+23.1)	61.52(+1.18)	95.57(↑ 0.12)	70.01(↑ 0.61)	77.52(↑ 0.30)	95.77	98.19	77.68	87.86	46.10	53.52	60.48	70.52
SegFormer+ [29]	104.7(+22.6)	3.82(+0.10)	95.29(↑ 0.06)	69.75(↑ 1.24)	78.30(↑ 1.86)	95.51	97.85	76.45	87.59	44.46	54.44	62.56	73.32
DPT+ [30]	235.5(+10.4)	111.6(+1.60)	94.30(↑ 0.46)	63.13(↑ 0.36)	74.98(↑ 0.08)	94.90	97.45	72.00	83.98	35.28	49.04	50.34	69.43

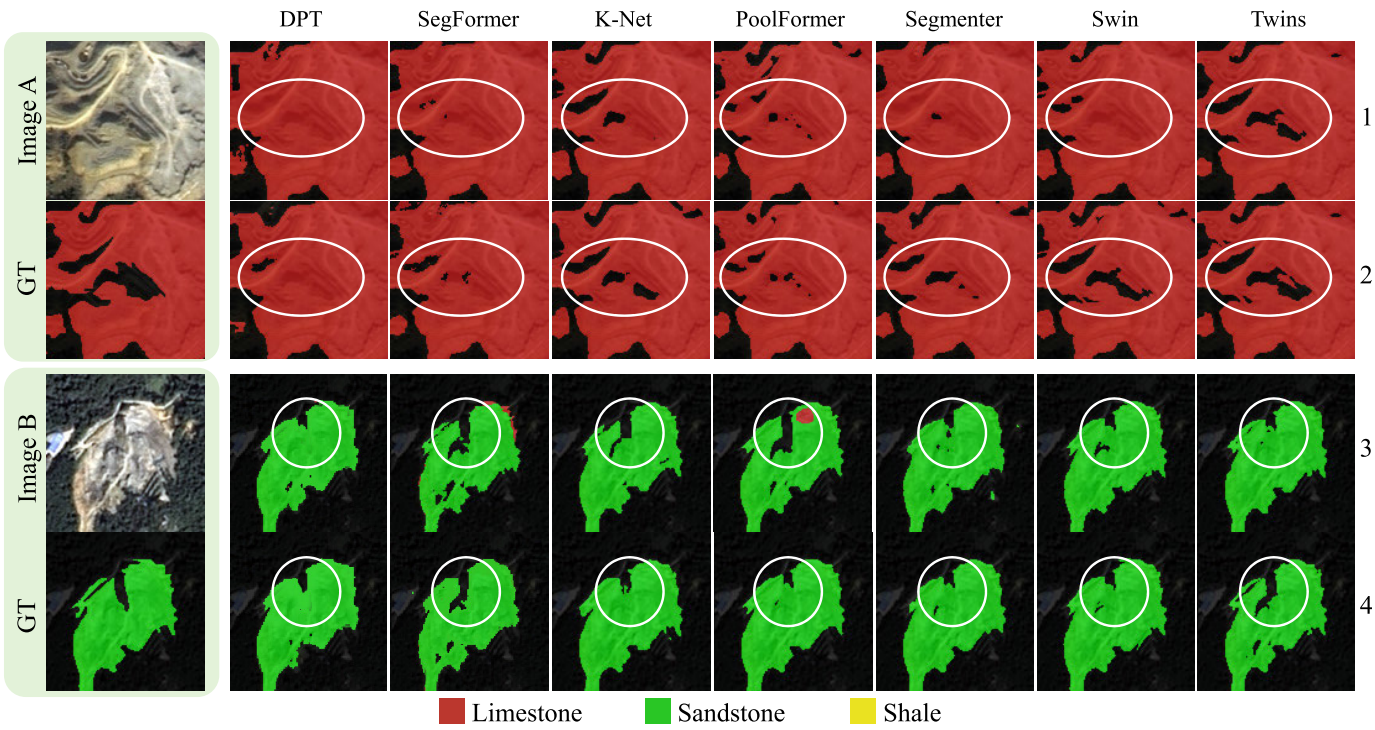


Fig. 5. Results on Fine-RSMI dataset. The first and third rows show the results of the corresponding baseline frameworks, which are the qualitative representations of Table IV. The second and fourth rows show the results after inserting EDFEM and ESM into the corresponding frameworks, which are the qualitative representations of Table V.

large open-pit mineral areas. However, with the assistance of our designed modules, the model pays more attention to both the internal and external edges of the mineral areas, effectively detecting nonmineral areas inside. Particularly, the segmentation results shown in the fourth row of image B are superior to the corresponding baseline results, especially in the detailed edge regions depicted in the images.

4) Visualization of Feature Distribution: To offer a more intuitive demonstration of the effectiveness of our proposed method in tackling the challenges posed by large intraclass variances and high interclass similarities, we randomly sample the test set and apply t-distributed stochastic neighborhood embedding (t-SNE) to map the high-dimensional features into a 2-D space. As shown in Fig. 6(a), the initial feature distribution indicates significant intraclass variances and high overlap among different classes. Combining the visualized segmentation results in Fig. 5, a comparison of the feature distribution plots in Fig. 6(b) and (c) reveals that our method alleviates the issue of misclassification of pixels within local regions near the edges. After inserting our designed modules, more feature points near the decision boundaries tend to cluster toward the feature centers. This effectively reclassifies the edge pixels, thereby enhancing the overall segmentation accuracy of the model.

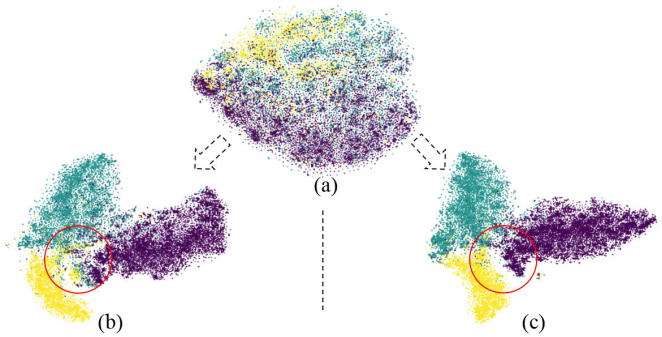


Fig. 6. Feature distribution maps of test set visualized using t-SNE. (a) Initial feature distribution. (b) Result after twins's baseline. (c) Result after inserting EDFEM and ESM.

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TABLE VI
RESULTS OF SOTAS WITH OUR MODULES ON WHDLD DATASET

Model	OA	mIoU	mAcc	Buiding		Road		Pavement		Vegetation		Bare Soil		Water	
				IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Twins [7]	86.20	66.16	77.62	62.13	73.99	64.92	77.23	47.21	**66.02**	**83.12**	91.63	43.85	59.31	95.72	97.52
Twins+	**86.36**(↑ 0.16)	**66.37**(↑ 0.21)	**78.64**(↑ 1.02)	**62.29**	77.71	**64.60**	**81.64**	**48.07**	65.33	82.98	90.35	**44.56**	59.12	95.70	**97.69**
Swin [26]	86.21	66.09	77.34	62.01	75.99	64.52	78.52	47.34	62.80	82.90	**92.00**	44.12	57.13	95.66	97.60
Swin+	86.24(↑ 0.03)	66.17(↑ 0.08)	77.71(↑ 0.37)	62.28	**78.14**	64.72	78.67	47.89	64.69	82.99	91.05	43.41	56.03	**95.71**	97.65
Segmenter [27]	83.79	62.37	75.80	56.27	72.87	60.55	80.26	42.14	57.84	79.99	88.69	41.41	58.29	93.84	96.85
Segmenter+	85.34(↑ 1.55)	64.49(↑ 2.12)	76.92(↑ 1.12)	59.88	74.13	61.85	79.97	44.86	59.40	82.00	91.21	43.26	**59.74**	95.11	97.09
PoolFormer [28]	85.73	64.94	76.54	61.00	76.98	62.17	76.78	46.05	61.91	82.54	91.30	42.44	54.79	95.46	97.47
PoolFormer+	85.85(↑ 0.12)	65.18(↑ 0.24)	76.68(↑ 0.14)	61.24	76.92	63.00	77.41	46.58	62.41	82.58	91.40	42.16	54.43	95.54	97.53
K-Net [32]	85.35	64.41	75.81	60.48	76.61	64.59	78.77	45.17	61.60	82.00	91.21	39.33	49.76	94.90	96.91
K-Net+	85.59(↑ 0.24)	64.81(↑ 0.40)	76.43(↑ 0.62)	60.66	76.81	63.86	78.64	46.03	62.87	82.38	90.99	40.59	52.03	95.35	97.23
SegFormer [29]	84.36	62.11	73.35	56.68	72.23	57.09	70.82	43.20	59.52	80.95	91.63	39.98	49.52	94.78	96.40
SegFormer+	84.50(↑ 0.14)	62.65(↑ 0.54)	74.76(↑ 1.41)	57.04	71.81	57.50	74.12	44.34	62.83	81.31	90.29	40.81	52.55	94.93	96.96
DPT [30]	82.64	59.12	70.81	52.85	72.18	51.73	64.49	40.15	56.76	79.29	89.85	37.84	45.92	92.83	95.66
DPT+	83.37(↑ 0.73)	60.67(↑ 1.55)	72.68(↑ 1.87)	54.87	72.64	54.67	68.80	41.48	57.93	79.98	90.02	39.67	50.65	93.36	96.03

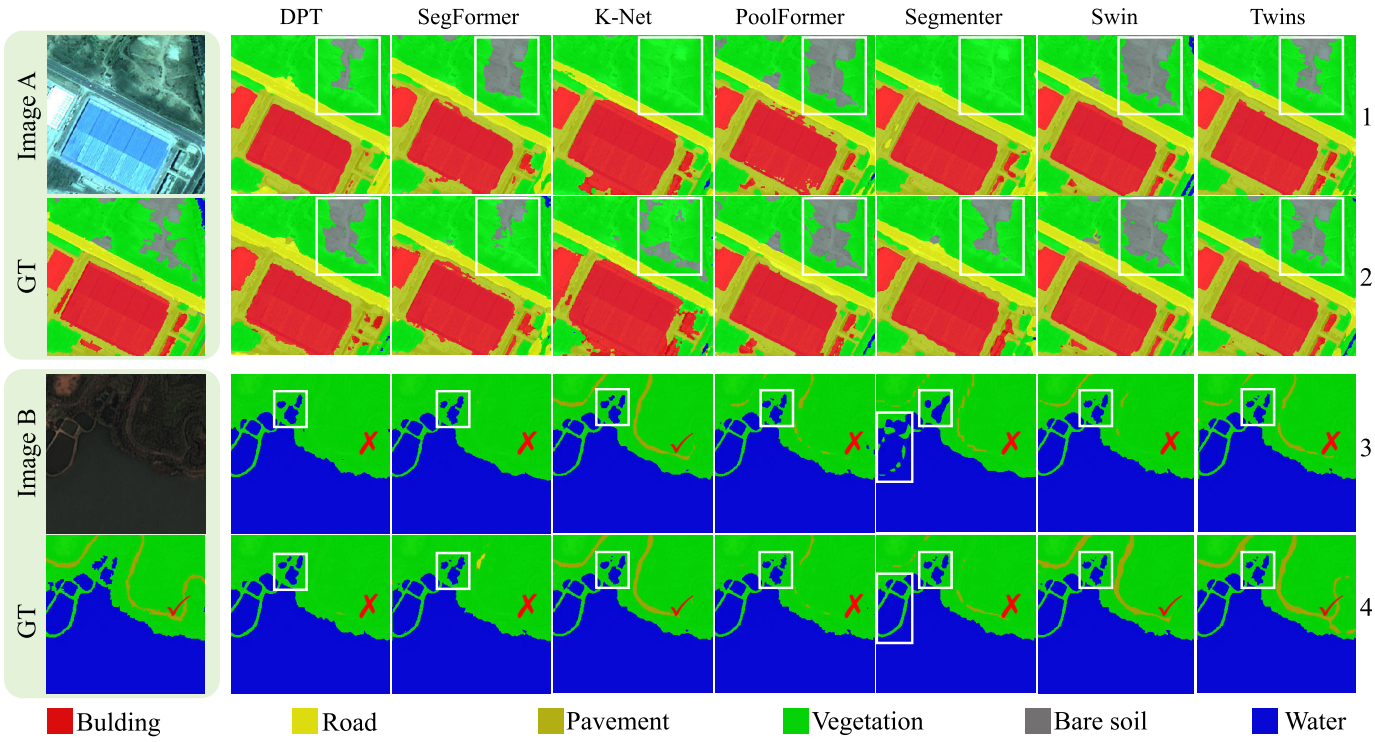


Fig. 7. Results on WHDLD dataset. The first and third rows show the results of the corresponding baseline frameworks. The second and fourth rows show the results after inserting our module into the corresponding frameworks.

D. Results on WHDLD Dataset

1) Main Results: Similarly, we insert the designed plug-and-play modules (EDFEM and ESM) into the baseline frameworks and evaluate its performance on the WHDLD [8] dataset. Based on the experimental data in Table VI, in terms of each class, the accuracy of the "Water" class is the highest, with both IoU and Acc around 95%. Conversely, the accuracy of the "Bare Soil" class is generally the lowest. As depicted in Image A of Fig. 7, the region of "Bare Soil" bears similarity to the mineral area type in fine-RSMI, exhibiting irregular, gradual transition edges that are prone to misclassification with surrounding environments. In terms of the morphology of segmented targets, the "Water" and "Bare Soil" classes possess the most irregular edges. Our modules are specifically designed to reinforce learning within the local pixel range of edges, hence, performing well on the "Water" and "Bare Soil" classes. Despite the already high accuracy of the "Water" class in the original baselines, further performance improvements are achieved after integrating our modules. Our framework is designed based on Twins [7], and as expected, the maximum mAcc is achieved with 78.64% on Twins+.

2) Efficiency Analysis: The results in Table VI show that the insertion of our modules in the seven frameworks achieves relative performance gains in three key metrics: OA, mIoU,

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TABLE VII
RESULTS OF SOTAS WITH OUR MODULES ON ISPRS DATASETS

Dataset	Model	OA	mIoU	mAcc	Imp.surf.		Building		Low veg.		Tree		Car	
					IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Vaihingen	Twins [7]	90.47	81.40	89.66	86.49	93.11	91.38	95.98	72.10	81.03	80.49	**92.22**	76.52	85.94
	Twins+	90.65(↑ 0.18)	82.02(↑ 0.62)	90.30(↑ 0.64)	**86.62**	**93.44**	**91.75**	**96.12**	**72.57**	81.79	**80.80**	91.70	**78.37**	**88.46**
	Swin [26]	90.27	81.03	89.57	85.93	92.85	91.61	95.98	71.34	80.88	80.22	91.65	76.05	86.49
	Swin+	90.31(↑ 0.04)	81.22(↑ 0.19)	89.59(↑ 0.02)	86.06	92.82	91.33	95.78	71.81	81.12	80.33	92.09	76.57	86.13
	Segmenter [27]	89.37	76.87	85.91	84.47	91.93	89.83	94.89	70.82	**82.28**	79.29	89.93	59.95	70.52
	Segmenter+	90.00(↑ 0.63)	78.95(↑ 2.08)	88.18(↑ 2.27)	85.27	92.94	90.08	95.06	72.24	82.25	80.26	90.44	66.88	80.21
	SegFormer [29]	89.31	78.27	87.71	83.72	91.88	89.42	94.79	70.39	80.59	79.94	90.93	67.87	80.35
	SegFormer+	89.51(↑ 0.20)	78.75(↑ 0.48)	88.43(↑ 0.72)	84.16	92.43	90.14	94.68	70.67	80.99	79.76	90.66	69.04	83.38
Potsdam	Twins [7]	91.26	86.40	93.13	**88.15**	94.35	**94.03**	97.49	78.08	90.22	80.09	87.59	91.63	95.98
	Twins+	91.34(↑ 0.08)	86.62(↑ 0.22)	93.18(↑ 0.05)	87.97	94.13	93.92	97.50	**78.43**	90.30	**80.63**	**88.56**	92.13	95.39
	Swin [26]	91.06	86.20	93.01	87.56	**94.48**	93.78	**97.68**	77.88	89.69	79.84	87.17	91.93	**96.01**
	Swin+	91.19(↑ 0.13)	86.47(↑ 0.27)	93.08(↑ 0.07)	87.69	93.68	93.86	97.63	78.12	**90.57**	80.36	87.91	**92.33**	95.60
	Segmenter [27]	89.26	83.21	91.37	85.32	92.24	91.27	95.57	74.42	88.19	76.43	87.27	88.61	93.56
	Segmenter+	89.84(↑ 0.58)	84.16(↑ 0.95)	91.95(↑ 0.58)	86.05	92.81	92.12	96.27	75.19	88.78	78.27	88.04	89.19	93.84
	SegFormer [29]	89.74	83.54	91.34	85.40	92.47	91.19	96.83	76.01	88.69	77.95	86.61	87.16	92.09
	SegFormer+	89.83(↑ 0.09)	83.94(↑ 0.40)	91.80(↑ 0.46)	85.53	92.38	91.20	97.08	76.00	88.39	77.93	87.11	89.02	94.04

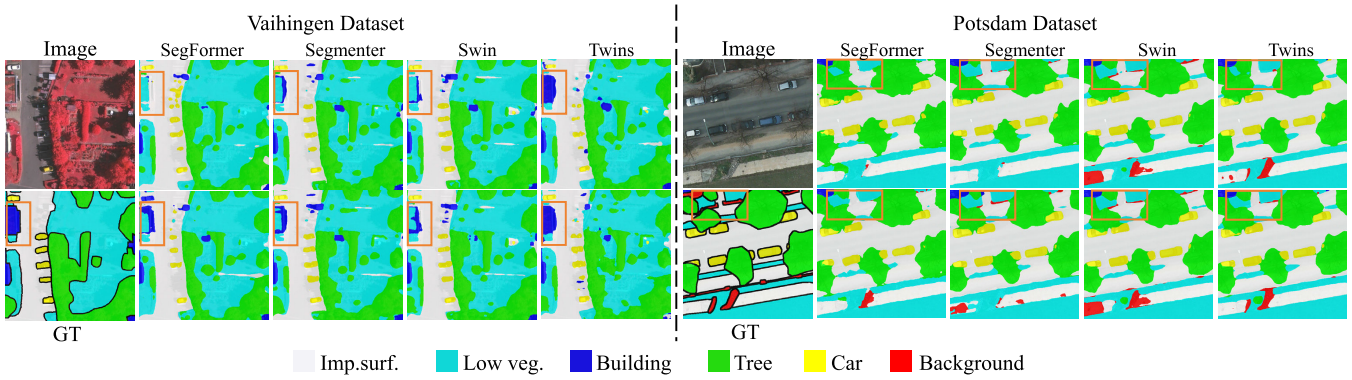


Fig. 8. Results on Vaihingen and Potsdam datasets. The first row show the results of the corresponding baseline frameworks. The second row show the results after inserting our module into the corresponding frameworks.

and mAcc. Particularly, the Segmenter+ framework demonstrates the most significant improvement, achieving a notable increase of 2.12% in mIoU. Extensive experiments validate that our designed plug-and-play modules enhance the performance of remote sensing images segmentation frameworks. This effectively demonstrates the generality and robustness of our proposed method.

3) Qualitative Results: The samples shown in Fig. 7 illustrate that our method excels in segmenting complex edges. After using our modules, the "Bare Soil" region shown in Image A and the "Water" region shown in Image B obtain results closer to the actual edge of ground truth. Furthermore, in the examples shown in the third and fourth rows, only K-Net, K-Net+, Swin+, and Twins+ frameworks accurately identify the elongated pavement parts, while other frameworks largely overlook these details. Among all the visualization results, Twins+ achieves the most ideal segmentation performance.

E. Results on ISPRS Dataset

1) Main Results: We implement a series of comparison experiments before and after incorporating the proposed modules on ISPRS dataset (Vaihingen and Potsdam). The results are shown in Table VII. The incremental values in parentheses of Table VII represent the quantitative improvement relative to the original baseline frameworks.

We select four baseline frameworks with the best performance: Twins [7], Swin [26], Segmenter [27], and SegFormer [29]. Twins+ also achieves the best performance on the Vaihingen and Potsdam datasets, with OA of 90.65% and 91.34%, respectively. These results nearly reach the current state-of-the-art performance on the ISPRS dataset by the DC-Swin [17] and UNetFormer [25].

2) Efficiency Analysis: In the results shown in Table VII, all baseline frameworks achieve stable performance improvements after incorporating our modules. As discussed in Section V-C, the additional time and parameter costs introduced by our designed modules are manageable. Based on this, our modules yield significant improvements across different algorithms on both the Vaihingen and Potsdam datasets. Our method performs as expected across different datasets and algorithms. The proposed modules not only enhances segmentation accuracy and robustness but also significantly simplifies the application process.

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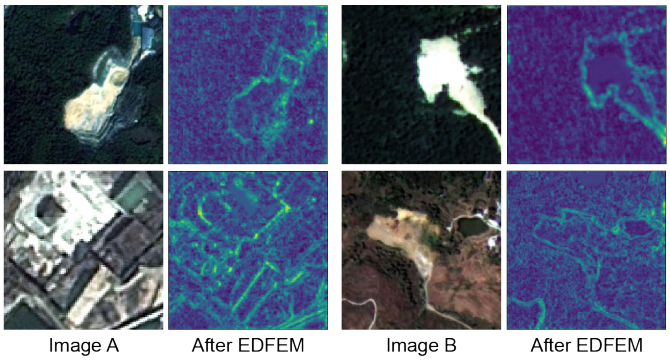


Fig. 9. Explicit edge enhanced visualization. Image A and B represent the input of the network, and the corresponding feature maps are the lowest-level features Edge_1 after being processed by EDFEM.

TABLE VIII
ABLATION STUDY OF PROPOSED MODULES ON FINE-RSMI

Feature Level 1	Feature Level 2	Feature Level 3	ESM	OA	mIoU	mAcc
✗	✗	✗	✗	95.92	73.19	80.45
✓	✗	✗	✗	**96.02**↑	73.76↑	80.32
✗	✓	✗	✗	95.90	73.42↑	81.14↑
✗	✗	✓	✗	96.01↑	73.60↑	80.48↑
✓	✓	✗	✗	95.87	73.19	81.11↑
✓	✗	✓	✗	95.89	72.55	80.41
✗	✓	✓	✗	95.95↑	73.68↑	80.88↑
✓	✓	✓	✗	95.98↑	73.93↑	**82.06**↑
✓	✓	✓	✓	**96.02**↑	**74.12**↑	81.57↑

3) Qualitative Results: We visualize some results from ISPRS dataset, with specific qualitative results shown in Fig. 8. In the Vaihingen dataset, our method generally performs well for the "Building" category. From the localized segmentation results within the bounding boxes in Fig. 8, it can be seen that the improved method outperforms the original baselines. The added module helps alleviate interclass ambiguity issues to some extent. In the Potsdam dataset, targets adjacent to the "Tree" category have complex edge shapes. Although the ground truth labels have relatively smooth contours, the actual edges of targets in the image are difficult to discern. As shown in the results in Fig. 8, our method improves the segmentation of complex edges of targets.

F. Ablation Study

To validate the effectiveness of our module designs and parameter settings, we apply Twins+ to perform ablation experiments on Fine-RSMI dataset.

1) Effect of Different Enhanced Feature Levels and Loss: Upon inserting the EDFEM into the backbone of Twins-PCPVT [7], we obtain multilevel feature maps with the same number and dimensions as the original features. The enhanced features consist of four levels denoted as $\boldsymbol{EF}_1$, $\boldsymbol{EF}_2$, $\boldsymbol{EF}_3$, and $\boldsymbol{EF}_4$. We systematically replace or combine features at different levels, as shown in Table VIII. $\boldsymbol{EF}_4$ is a simple lateral connection and not considered as an enhanced feature. The symbol '↑' indicates an improvement in the original result compared to the first row in the table without replacing any feature layers. Results demonstrate that replacing all features with enhanced features $\boldsymbol{EF}_1$, $\boldsymbol{EF}_2$, and $\boldsymbol{EF}_3$ achieves the best performance, with a peak mAcc of 82.06%.

TABLE IX
ABLATION EXPERIMENTS WITH DIFFERENT COMBINATIONS OF COEFFICIENTS α AND β ON FINE-RSMI

α	β	OA	mIoU	mAcc
0.3	0.5	95.93	73.28	80.02
0.5	0.3	95.90	73.31	81.12
0.5	0.5	95.96	73.71	**81.57**
0.5	**0.8**	**96.02**	**74.12**	**81.57**
0.8	0.5	95.96	73.69	81.11
1.0	1.0	95.98	74.06	81.23

TABLE X
ABLATION EXPERIMENTS WITH DIFFERENT COMBINATIONS OF COEFFICIENTS α AND β ON WHDLD

α	β	OA	mIoU	mAcc
0.3	0.5	86.20	**66.37**	78.62
0.5	0.3	86.23	66.10	77.89
0.5	0.5	86.23	66.23	78.63
0.5	**0.8**	**86.36**	**66.37**	**78.64**
0.8	0.5	**86.36**	66.27	78.63
1.0	1.0	86.25	66.34	78.28

We visualize the feature maps after EDFEM, as shown in Fig. 9. The blue highlighted regions represent the edges of mineral areas, with a width greater than one pixel, which is the result of local dilation of edges after fusion. By examining the feature maps after EDFEM, it is demonstrated that our modules explicitly enhances attention within the local range of mineral area edges, attenuating attention on smooth interiors of the mineral areas and other background regions. This effectively filters out irrelevant information unrelated to edges and facilitates guiding the network to focus on learning mineral area edges.

2) Effect of Auxiliary Classification Head: The experimental results in Table VIII demonstrate that when only replacing a single layer to enhance features, the overall network segmentation accuracy also improves to varying degrees. Among them, enhancing the features of the lowest layer has the most significant effect, with an mIoU of 73.76%. The lowest layer features contain richer detail texture information. Therefore, we use $\boldsymbol{Edge}_1$ (the output feature of $\boldsymbol{EF}_1$) as the input for the auxiliary classification head to calculate the edge supervision loss. As expected, further performance improvement is achieved when supervised training is conducted using the ground truth edge maps. That is, after adding ESM, the OA reaches the highest at 96.02%, and the mIoU reaches the highest at 74.12%.

3) Impact of α and β Coefficients: We discuss the selection of edge weight coefficient α and feature weight coefficient β through experiments. Proper balancing of these parameters is beneficial for supplementing detailed information of the underlying features. As shown in Tables IX and X, different combinations have varying degrees of impact on mIoU and mAcc of the test set. Our model achieves optimal performance on both the Fine-RSMI and WHDLD datasets when $\alpha = 0.5$ and $\beta = 0.8$. Therefore, we choose $\alpha = 0.5$ and $\beta = 0.8$ as the parameter configuration for the Twins+ framework.

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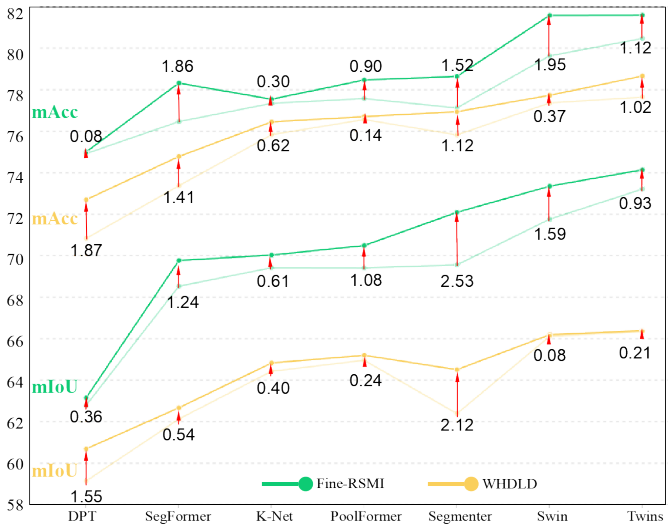


Fig. 10. Quantitative representation of plug-and-plug capability on Fine-RSMI and WHDLD. Light-colored lines represent the accuracy of baselines, darker colored lines represent the accuracy after integrating EDFEM and ESM. The incremental values in the graph represent the improvement of our frameworks relative to the baselines.

Coefficients α and β are used to balance different components of the EDFEM. Specifically, α adjusts the edge feature map, while β adjusts the supplementary detail information map. To ensure the robustness of the chosen coefficients, we conduct cross-validation on multiple datasets, confirming their effectiveness in achieving optimal model performance. Furthermore, we perform a preliminary sensitivity analysis using a parameter grid search method to study the impact of different coefficient values on model performance, further validating our choices.

4) Plug-and-Plug Capability: As shown in Tables V and VI, we respectively report the results of applying our plug-and-play modules (EDFEM and ESM) to the corresponding baseline frameworks on Fine-RSMI and WHDLD. Our designed modules can be seamlessly integrated into the backbone of frameworks with hierarchical features, thus augmenting the segmentation network's capacity to learn intricate image edge details. As depicted in Fig. 10, this plugin demonstrates significant gains across different frameworks. And this plugin further enhances segmentation accuracy across different remote sensing datasets.

VI. Conclusion

We establish a fine-annotated benchmark dataset of remote sensing mineral images, providing a solid foundation for research in relevant fields. Furthermore, we propose two collaborative modules: EDFEM fine-tunes the feature representation of mineral edges, and ESM guides the network to enhance the learning of detailed information at the edges for more precise edge segmentation. Our proposed method operates in a plug-and-play manner, greatly enhancing its flexibility and practicality, and has been proven effective across a range of frameworks. Additionally, quantitative and qualitative results conclusively demonstrate that our proposed method attains state-of-the-art performance.

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